Supplementary Materials: Explainable anomaly detection for an industrial air compressor: temporal evaluation and training-data contamination on the MetroPT-3 dataset

This document accompanies the main paper submitted to ICASIM 2026. Section, table, figure and equation labels carry the prefix S (for example, Table S1). Everything referred to as "Supplementary" in the main paper is defined here. The code, configuration files, per-run results and data tables behind every item are in the public repository named in the data availability statement of the main paper.

S1. Data, signals and labeling

MetroPT-3 contains 15 signals (Table S1) from a metro-train air production unit. We used a partial copy: 436,151 of 1,516,948 rows (28.8 %), covering 1–4 February and 1 May to 26 June 2020 at a median sampling interval of 10 s. The supplied rows have no missing values, duplicates or out-of-order timestamps, and 89 gaps longer than 60 s divide them into 90 contiguous segments. Documented failures 1 (18 April) and 4 (15 July) have no rows in this copy.

Table S1. Signals of the MetroPT-3 air production unit as named in the data files, with the sensor groups used in the ablations and the window features computed from each.

Signal	Type	Group	Window features
TP2, TP3, H1, DV_pressure, Reservoirs	analog (pressure)	pressure	mean, std, min, max, delta
Oil_temperature, Motor_current	analog	thermal/current	mean, std, min, max, delta
COMP, DV_eletric, Towers, MPG, LPS, Pressure_switch, Oil_level, Caudal_impulses	digital	digital	duty cycle, transition count

The dataset has no per-observation labels. Windows are assigned to the zones event, pre-onset (24 h before onset), post (event end to maintenance) and normal from the failure reports, and windows with Oil_level ≠ 1 or Caudal_impulses ≠ 1 receive an additional state flag. Table S2 lists the splits, event intervals and zones. The report table numbers two rows "#1", and the remark for failure 2 reads "30Apr", which we interpret as 30 May 12:00 (an inference). Fig. S1 shows the signal distributions and the share of loaded-compressor samples; Fig. S2 shows the sensor traces around failure 3 and the flat-lined signals on 12 June; Fig. S3 places both evaluable failures against the data coverage.

Table S2. Chronological splits, documented failure events and ambiguity zones used in the evaluation.

Item	Definition
Training	2020-02-01 to 02-04 and 05-01 to 05-21 (windows fully inside); 5,710 windows
Validation	2020-05-22 to 05-28 23:30 (threshold calibration, clean windows only); 1,414 windows, 409 flagged
Test	2020-05-28 23:30 to 2020-06-26; 6,962 windows (644 event, 4,933 normal, 766 flagged, 522 pre-onset, 97 post)
Failure 2	05-29 23:30 – 05-30 06:00; maintenance 05-30 12:00 (inferred from the remark "30Apr")
Failure 3	06-05 10:00 – 06-07 14:30; maintenance 06-08 (data resume 11:48)
Failures 1 and 4	No rows in the supplied data; not evaluated
Pre-onset / post	24 h before onset; event end to maintenance. Excluded from AUROC and false alarms

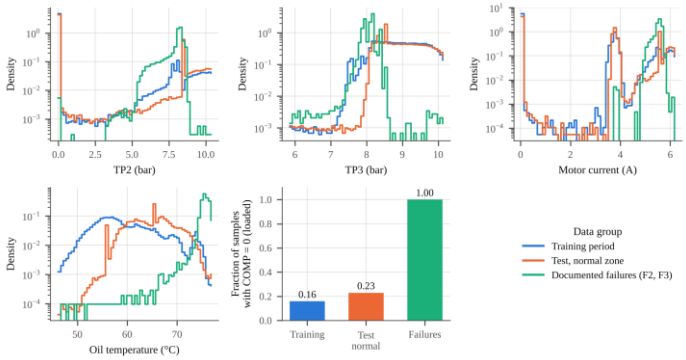


Figure S1. Signal distributions and compressor load: histograms of the analog signals and the fraction of samples with COMP = 0 (loaded compressor) in the training, normal-test and failure periods.

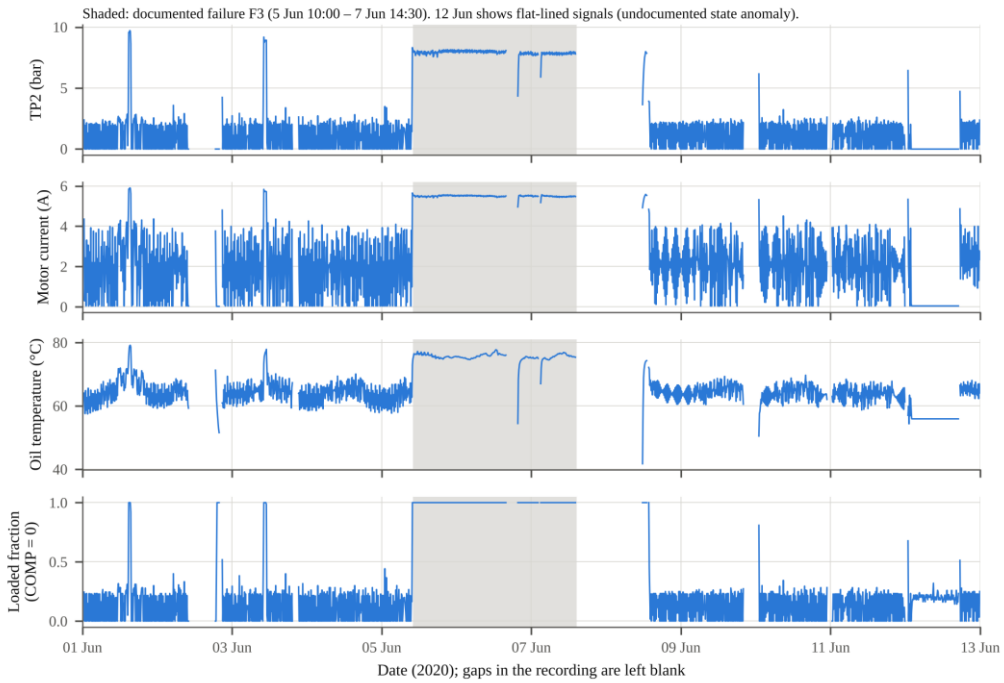


Figure S2. Sensor time series, 1–13 June. Shaded: documented failure 3 (5 June 10:00 – 7 June 14:30). The signals are flat-lined on 12 June, an undocumented state anomaly.

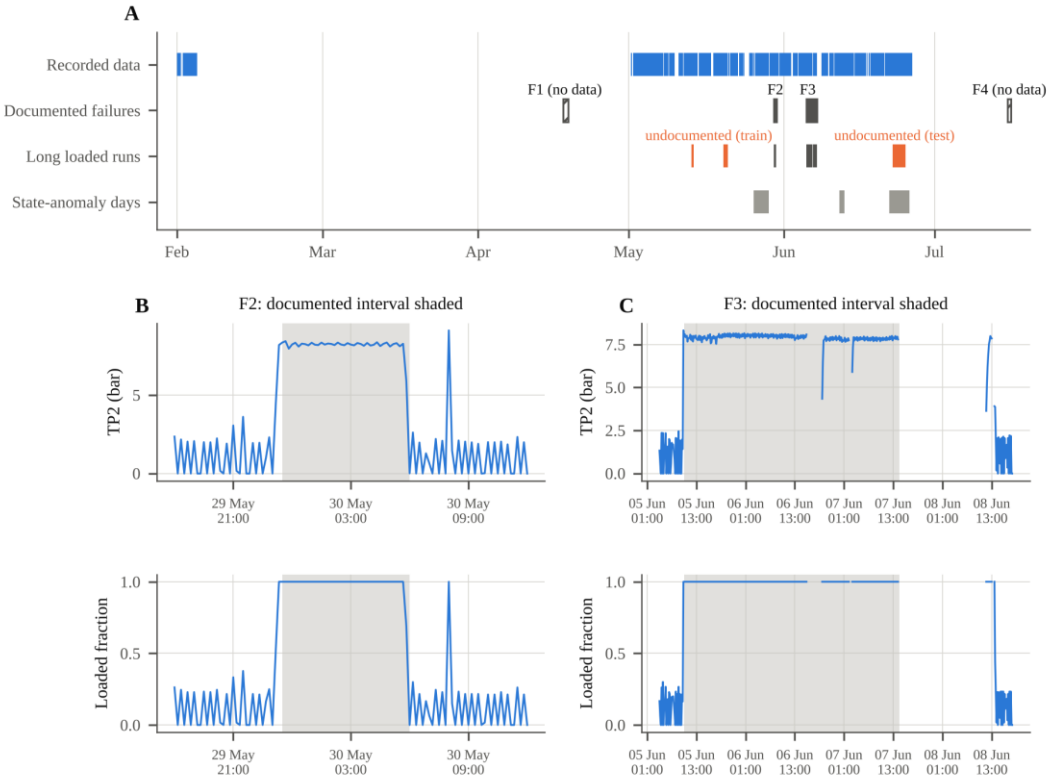


Figure S3. Documented failure intervals against data coverage (top), and sensor traces around failures 2 and 3 with the documented interval shaded (bottom panels).

S2. Preprocessing and implementation details

Within each contiguous segment we cut causal windows of W = 180 samples (30 min) at a stride of 30 samples (5 min). A window is stamped with the time of its last sample and never crosses a gap longer than 60 s. For analog signal j the features are mean, standard deviation, minimum, maximum and last-minus-first difference, and for digital signals the duty cycle and the number of transitions (51 candidate features, Table S1):

$$x_w = [\varphi(s_{j,t}),\ t \in (t_w - W\Delta,\ t_w]] \tag{S1}$$

Features with a training standard deviation below 0.01 are removed (5 of 51; 46 remain), and the remaining features are standardized with training statistics only:

$$z_{wk} = (x_{wk} - \mu_k) / \sigma_k,\ \ \mu_k, \sigma_k \text{ from training windows} \tag{S2}$$

The LSTM autoencoder reads each window as 1-minute block means (5 steps × 15 signals). The pipeline is summarized in Fig. S4, and the detector configurations are in Table S3. Hyper-parameters were fixed in advance and were not tuned on the test period; validation data were used only for threshold calibration and for choosing the autoencoder epoch.

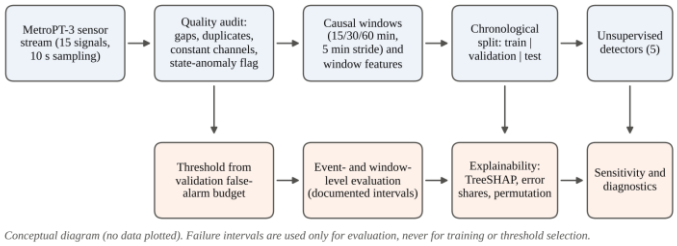


Figure S4. Overview of the evaluation pipeline: chronological splits, causal windows, train-only preprocessing, validation-fixed thresholds, test-time scoring and the explanation analyses.

Table S3. Detector configurations (from the configuration files and model code shipped with the repository).

Model	Input	Configuration	Score	Seeds
Rolling z-score	12 mean/duty features	training mean and std	max |z|	1
Isolation Forest	46 features	scikit-learn, 200 trees, random_state = seed	mean path length	0–4
One-class SVM	46 features	RBF, gamma = scale, nu = 0.05, ≤ 4,000 random training windows	decision value	0
MLP autoencoder	46 features	46-32-8-32-46, ReLU, Adam 1e-3, weight decay 1e-5, batch 128, 150 epochs, best epoch by validation MSE	reconstruction MSE	0–4
LSTM autoencoder	5×15 block means	1-layer LSTM encoder/decoder, hidden 32, Adam 2e-3, batch 128, 40 epochs, best epoch by validation MSE	reconstruction MSE	0–2

S3. Evaluation protocol and metrics

An alert is raised when a window's score reaches the threshold τ, set so that clean validation windows (22–28 May without the state flag; 1,005 windows) yield at most one alert episode per 24 h (main paper, Eq. 1). Alert windows less than 60 min apart form one episode. Budgets of 0.25, 0.5, 1, 2 and 4 episodes per 24 h are used for the sensitivity analysis. The false-alarm rate on the test period is

$$FA_{24} = 24 \cdot E_{test} / H_{normal} \tag{S3}$$

where E_{test} is the number of alert episodes that start in normal windows and H_{normal} the normal test time in hours (number of normal windows × 5 min). Pre-onset and post windows are excluded from AUROC, AP and false-alarm counts. AUROC and AP compare event windows with normal windows; confidence intervals resample whole calendar days with replacement (day-block bootstrap, 1,000 resamples). The audit protocol removes the 766 state-flagged test windows from the normal class, whereas the strict protocol keeps them. An event is detected if any alert falls inside the documented interval; delay is the first alert minus the documented onset.

S4. Extended window-level and event-level results

Table S4 gives the window-level results with bootstrap intervals for AUROC and AP under both protocols, and Fig. S5 plots them. Event-level detection, delays and pre-onset alerts are in Table S5; false-alarm rates and the alerted share of

state windows are in Table S6, and Fig. S6 summarizes both. Bootstrap intervals resample days of a single test period and are optimistic; the near-zero seed spread of Isolation Forest reflects only its internal randomness.

Table S4. Window-level test results (30-min windows, primary setting). Mean ± SD over seeds; interval = 95 % day-block bootstrap (seed 0).

Model	Seeds	AUROC	AUROC 95 % CI	AP	AP 95 % CI	AUROC (strict)	AP (strict)
Rolling z-score	1	0.995	[0.990, 0.999]	0.880	[0.62, 0.97]	0.889	0.314
Isolation Forest	5	0.991 ± 0.000	[0.982, 0.998]	0.824 ± 0.005	[0.48, 0.95]	0.861 ± 0.001	0.273 ± 0.002
One-Class SVM	1	0.873	[0.825, 0.973]	0.352	[0.12, 0.57]	0.755	0.170
MLP autoencoder	5	0.340 ± 0.087	[0.253, 0.454]	0.099 ± 0.010	[0.03, 0.19]	0.294 ± 0.075	0.067 ± 0.007
LSTM autoencoder	3	0.528 ± 0.109	[0.474, 0.622]	0.151 ± 0.068	[0.04, 0.28]	0.461 ± 0.096	0.101 ± 0.023

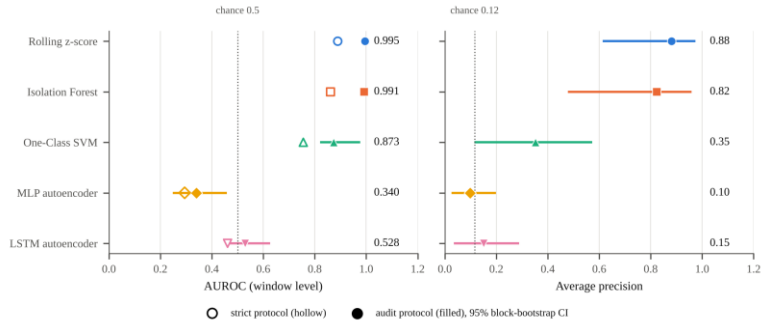


Figure S5. Window-level AUROC and AP of the five detectors on the test period with day-block bootstrap intervals; dotted lines mark the reference levels (chance AUROC; AP at event prevalence).

Table S5. Event-level detection at the primary budget. det. = detected / runs; delay = first alert minus documented onset (min); frac. = share of event windows in alert; pre24 = alerted windows in the 24 h pre-onset zone (seed mean).

Model	Seeds	F2 det.	F2 delay	F2 frac.	F2 pre24	F3 det.	F3 delay	F3 frac.	F3 pre24
Rolling z-score	1	1/1	2.3	1.00	0.0	1/1	8.0	1.00	0.0
Isolation Forest	5	5/5	2.3	0.48 ± 0.34	1.0	5/5	3.0	0.08 ± 0.05	0.2
One-Class SVM	1	0/1	–	0.00	0.0	1/1	2051.9	0.01	0.0
MLP autoencoder	5	5/5	2.3	0.04 ± 0.02	0.4	5/5	17.9	0.02 ± 0.00	0.0
LSTM autoencoder	3	0/3	–	0.00 ± 0.00	0.0	3/3	2051.9	0.01 ± 0.00	0.0

Table S6. False alarms at the primary budget: alert episodes per 24 h of normal test time under the audit and strict protocols, share of flagged digital-state windows in alert, and the validation false-alarm rate used for calibration.

Model	FA/24 h audit	FA/24 h strict	State windows alerted	Validation FA/24 h
Rolling z-score	0.88	0.76	0.80	0.86
Isolation Forest	0.47 ± 0.00	0.51 ± 0.00	0.99 ± 0.00	0.86 ± 0.00
One-Class SVM	0.76	0.76	1.00	0.86
MLP autoencoder	0.82 ± 0.34	0.77 ± 0.28	1.00 ± 0.00	0.86 ± 0.00
LSTM autoencoder	0.66 ± 0.03	0.91 ± 0.48	0.61 ± 0.33	0.76 ± 0.17

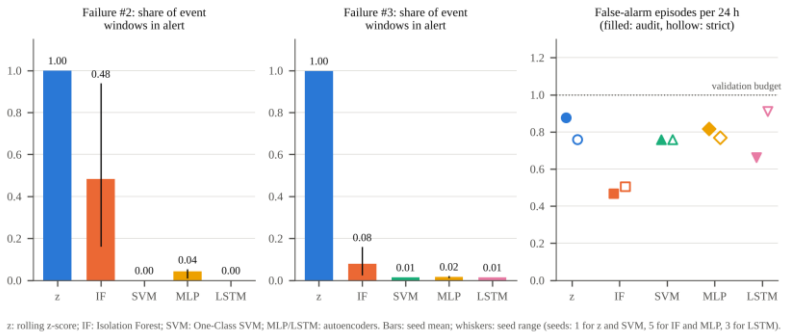


Figure S6. Event-level results per model: share of event windows in alert for each failure (bars: seed mean; whiskers: seed range) and false-alarm episodes per 24 h against the validation budget.

S5. Threshold sensitivity

Fig. S7 and Table S7 report detection and false alarms for budgets of 0.25–4 episodes per 24 h. At the loosest budget the z-score and Isolation Forest alert on many windows in the 24 h before onset, with first alerts up to about 24 h ahead, but at test false-alarm rates of 1.7–2.6 per 24 h. These alerts cannot be separated from ordinary false alarms or from an unrecognized regime change in that zone, so we draw no lead-time conclusion from them.

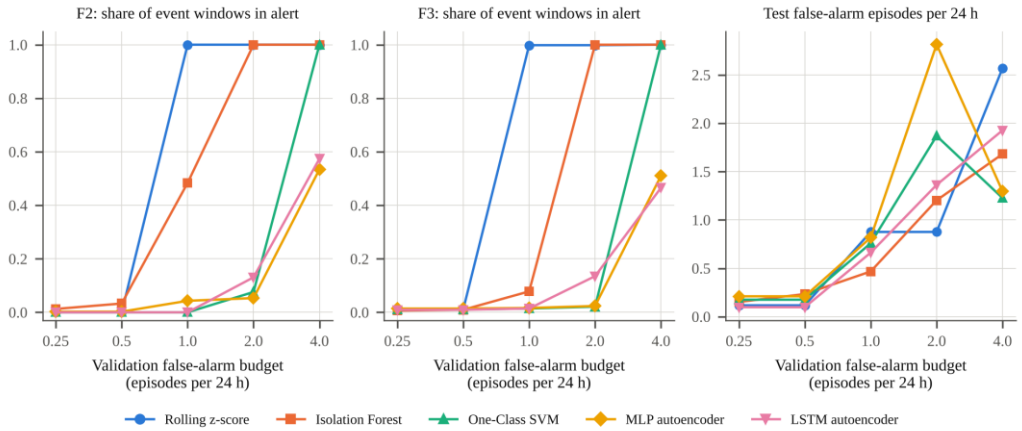


Figure S7. Sensitivity of detection and false alarms to the validation alert budget (0.25–4 episodes per 24 h). The primary budget is one.

Table S7. Threshold sensitivity: share of event windows in alert (F2, F3), events detected, and test and validation false-alarm episodes per 24 h for each budget.

Model	Budget/24 h	F2 frac.	F3 frac.	F2 det.	F3 det.	Test FA/24 h	Val. FA/24 h
Rolling z-score	0.25	0.000	0.009	0/1	1/1	0.12	0.00
Rolling z-score	0.50	0.000	0.011	0/1	1/1	0.12	0.29
Rolling z-score	1.00	1.000	0.998	1/1	1/1	0.88	0.86
Rolling z-score	2.00	1.000	0.998	1/1	1/1	0.88	1.72
Rolling z-score	4.00	1.000	1.000	1/1	1/1	2.57	2.29
Isolation Forest	0.25	0.013	0.006	3/5	5/5	0.15	0.00
Isolation Forest	0.50	0.033	0.010	4/5	5/5	0.23	0.29
Isolation Forest	1.00	0.484	0.078	5/5	5/5	0.47	0.86
Isolation Forest	2.00	1.000	1.000	5/5	5/5	1.20	1.72
Isolation Forest	4.00	1.000	1.000	5/5	5/5	1.68	3.61
One-Class SVM	0.25	0.000	0.011	0/1	1/1	0.18	0.00
One-Class SVM	0.50	0.000	0.011	0/1	1/1	0.18	0.29
One-Class SVM	1.00	0.000	0.014	0/1	1/1	0.76	0.86
One-Class SVM	2.00	0.076	0.021	1/1	1/1	1.87	1.72
One-Class SVM	4.00	1.000	1.000	1/1	1/1	1.23	2.87
MLP autoencoder	0.25	0.003	0.014	1/5	5/5	0.21	0.29
MLP autoencoder	0.50	0.003	0.014	1/5	5/5	0.21	0.29
MLP autoencoder	1.00	0.043	0.016	5/5	5/5	0.82	0.86
MLP autoencoder	2.00	0.053	0.024	5/5	5/5	2.81	1.72
MLP autoencoder	4.00	0.534	0.510	5/5	5/5	1.30	2.92
LSTM autoencoder	0.25	0.000	0.009	0/3	3/3	0.10	0.29
LSTM autoencoder	0.50	0.000	0.009	0/3	3/3	0.10	0.29
LSTM autoencoder	1.00	0.000	0.014	0/3	3/3	0.66	0.76
LSTM autoencoder	2.00	0.131	0.135	1/3	3/3	1.36	1.72
LSTM autoencoder	4.00	0.574	0.465	3/3	3/3	1.93	2.20

S6. Temporal generalization and ablations

Fig. S8 shows the daily share of windows in alert, the drift of the daily median oil temperature, and test AUROC by training period and window length. Table S8 lists every experiment: training on February only or May only, 15- and 60-minute windows, removal of the near-constant-feature filter, single sensor groups and group removal, and the contamination diagnostic. Training on four February days gave the highest autoencoder AUROC, and May alone the lowest; Section S7 gives the explanation. Each ablation was run with three seeds.

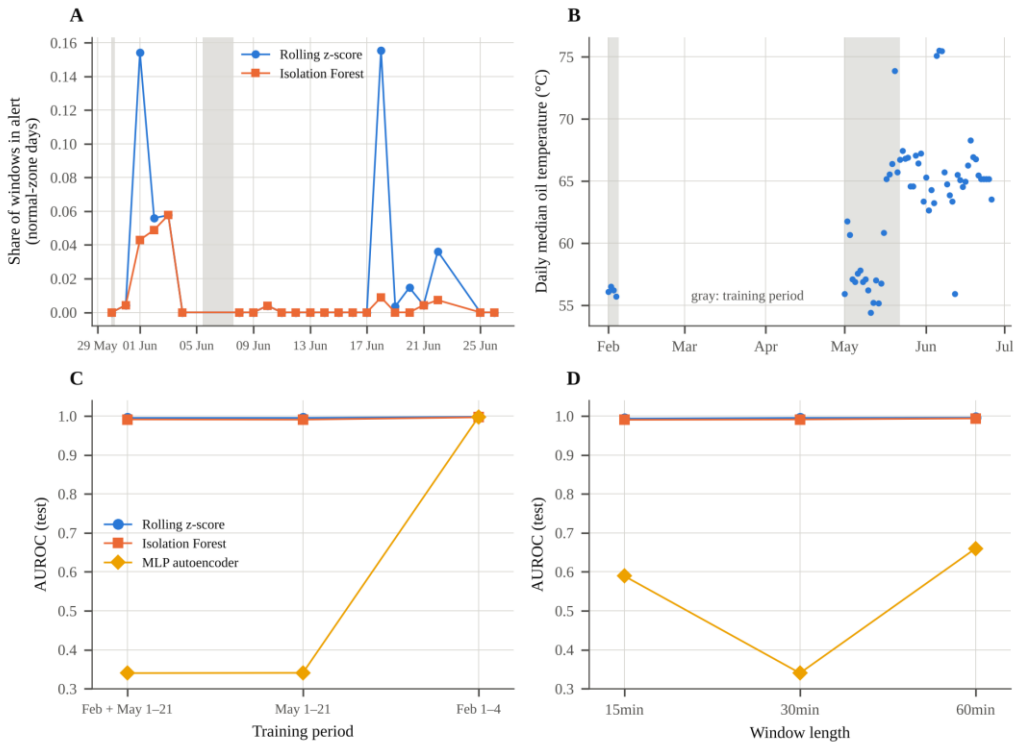


Figure S8. Temporal behavior: share of windows in alert per day, drift of the daily median oil temperature across the recording, and test AUROC by training period and by window length.

Table S8. All experiments: main setting, ablations and the contamination diagnostic (test period, primary budget). Mean over seeds; the z-score is deterministic.

Experiment	Setting	Model	n	AUROC	AP	F2 frac.	F3 frac.	FA/24 h
Main	Feb + May 1–21, 30 min	Rolling z-score	1	0.995	0.880	1.00	1.00	0.88
Main	Feb + May 1–21, 30 min	Isolation Forest	5	0.991	0.824	0.48	0.08	0.47
Main	Feb + May 1–21, 30 min	One-Class SVM	1	0.873	0.352	0.00	0.01	0.76
Main	Feb + May 1–21, 30 min	MLP autoencoder	5	0.340	0.099	0.04	0.02	0.82
Main	Feb + May 1–21, 30 min	LSTM autoencoder	3	0.528	0.151	0.00	0.01	0.66
Window length	15 min	Rolling z-score	1	0.993	0.849	1.00	1.00	0.69
Window length	15 min	Isolation Forest	3	0.990	0.808	0.08	0.04	0.55
Window length	15 min	MLP autoencoder	3	0.589	0.141	0.01	0.02	0.99
Window length	60 min	Rolling z-score	1	0.996	0.938	0.96	0.99	0.72
Window length	60 min	Isolation Forest	3	0.994	0.877	1.00	1.00	0.68
Window length	60 min	MLP autoencoder	3	0.660	0.239	0.06	0.02	0.70
Training period	May 1–21 only	Rolling z-score	1	0.995	0.880	0.99	1.00	0.88
Training period	May 1–21 only	Isolation Forest	3	0.991	0.816	0.61	0.05	0.53
Training period	May 1–21 only	MLP autoencoder	3	0.341	0.097	0.03	0.01	0.86
Training period	Feb 1–4 only	Rolling z-score	1	0.998	0.975	1.00	1.00	0.23
Training period	Feb 1–4 only	Isolation Forest	3	0.997	0.944	0.97	0.98	1.01
Training period	Feb 1–4 only	MLP autoencoder	3	0.997	0.928	0.98	1.00	0.76
Feature filter	no near-constant-feature filter	Isolation Forest	3	0.991	0.823	0.71	0.20	0.49
Feature filter	no near-constant-feature filter	MLP autoencoder	3	0.477	0.123	0.02	0.01	0.72
Sensor group	only pressure	Isolation Forest	3	0.992	0.835	1.00	1.00	0.64
Sensor group	only pressure	MLP autoencoder	3	0.899	0.405	0.05	0.01	1.23
Sensor group	only oil T + current	Isolation Forest	3	0.992	0.838	1.00	0.97	1.03
Sensor group	only oil T + current	MLP autoencoder	3	0.523	0.116	0.00	0.01	0.72
Sensor group	only digital	Isolation Forest	3	0.957	0.589	0.00	0.00	0.29
Sensor group	only digital	MLP autoencoder	3	0.742	0.393	0.00	0.01	0.86
Sensor group	without pressure	Isolation Forest	3	0.979	0.692	0.05	0.01	0.39
Sensor group	without pressure	MLP autoencoder	3	0.329	0.119	0.04	0.02	1.23
Sensor group	without oil T + current	Isolation Forest	3	0.991	0.817	0.15	0.03	0.47
Sensor group	without oil T + current	MLP autoencoder	3	0.817	0.329	0.03	0.02	0.64
Sensor group	without digital	Isolation Forest	3	0.992	0.836	1.00	0.97	0.64
Sensor group	without digital	MLP autoencoder	3	0.400	0.100	0.05	0.01	1.36
DIAGNOSTIC (not deployable)	undocumented loaded runs removed from training	Rolling z-score	1	1.000	0.990	1.00	1.00	0.23
DIAGNOSTIC (not deployable)	undocumented loaded runs removed from training	Isolation Forest	5	0.999	0.984	1.00	1.00	0.46
DIAGNOSTIC (not deployable)	undocumented loaded runs removed from training	One-Class SVM	1	1.000	1.000	1.00	0.99	0.53
DIAGNOSTIC (not deployable)	undocumented loaded runs removed from training	MLP autoencoder	5	1.000	1.000	0.99	1.00	0.81

DIAGNOSTIC (not deployable)	undocumented loaded runs removed from training	LSTM autoencoder	3	0.997	0.924	0.34	0.59	0.51

S7. Training-data contamination

Table S9 lists the seven loaded-compressor runs of 100 min or more. Four overlap documented failures; two lie in the training period (13–14 May and 19–20 May; 32.6 h in total) and one in the test period (22–25 June; 62.0 h). The onset offsets show that the sensor regime of failures 2 and 3 starts 15 and 12 min before the documented onset. Table S10 gives the results of retraining after excluding training windows that overlap the two undocumented training runs. The exclusion rule was formulated after the test results were seen, so this is a diagnostic only.

Table S9. Long loaded-compressor runs (COMP = 0 for at least 100 min). Medians of motor current (A), TP2 (bar) and oil temperature (°C); onset/end offsets are relative to the documented interval (min; negative = earlier).

Start	End	Min	Period	Documented	Motor A	TP2	Oil T	Onset offset	End offset
2020-05-13 13:44	2020-05-14 00:56	673	train	none	5.37	7.68	69.2	–	–
2020-05-19 22:22	2020-05-20 19:48	1286	train	none	5.51	8.06	73.8	–	–
2020-05-29 23:14	2020-05-30 05:56	402	test	F2	5.62	8.56	76.0	-15	-3
2020-06-05 09:48	2020-06-06 15:55	1807	test	F3	5.54	8.28	75.7	-12	-1355
2020-06-06 19:42	2020-06-07 02:13	391	test	F3	5.53	8.07	75.1	2022	-736
2020-06-07 03:05	2020-06-07 14:19	674	test	F3	5.51	8.11	75.6	2466	-10
2020-06-22 15:05	2020-06-25 05:08	3723	test	none	5.58	8.39	65.1	–	–

Table S10. Contamination diagnostic: models retrained after removing training windows that overlap the undocumented long loaded runs (post hoc; not a deployable pipeline). Mean over seeds; detected = events detected / runs.

Model	Runs	AUROC	AP	AUROC (strict)	FA/24 h	F2 detected	F3 detected
IsolationForest	5	0.999	0.984	0.998	0.46	5/5	5/5
LSTM-AE	3	0.997	0.924	0.929	0.51	2/3	3/3
MLP-AE	5	1.000	1.000	0.867	0.81	5/5	5/5
OneClassSVM	1	1.000	1.000	0.867	0.53	1/1	1/1
ZScore	1	1.000	0.990	1.000	0.23	1/1	1/1

S8. Explainability details

Table S11 lists the Isolation Forest TreeSHAP contributions and the z-score driver shares by sensor and zone; the explanation panels are Fig. 2 of the main paper. Permutation importance is in Table S12 (post hoc on the test period, descriptive only), and ranking stability in Table S13. The MLP autoencoder's explanations are not meaningful because its AUROC is below chance. All explanations describe model scores, not physical causes.

Table S11. Mean signed TreeSHAP contribution of each sensor to the Isolation Forest anomaly score (anomaly direction) by zone, and the share of the z-score's maximal feature by sensor.

Sensor	IF normal	IF F2	IF F3	IF state	z-score F2	z-score F3
TP2	0.03	0.85	0.77	1.09	1.00	0.79
TP3	0.07	0.73	0.67	0.91	0.00	0.00
H1	0.03	0.79	0.77	0.97	0.00	0.01
DV_pressure	-0.01	0.72	0.70	0.06	0.00	0.14
Reservoirs	0.07	0.68	0.65	0.85	0.00	0.00
Oil_temperature	0.36	0.49	0.45	0.31	0.00	0.00
Motor_current	0.14	0.67	0.59	0.83	0.00	0.00
COMP	0.07	0.37	0.35	0.43	–	–
DV_eletric	0.05	0.33	0.31	0.51	–	–
Towers	0.03	0.57	0.54	0.56	0.00	0.04
MPG	0.06	0.45	0.43	0.54	–	–
LPS	-0.01	-0.01	0.00	-0.01	0.00	0.01
Pressure_switch	0.02	0.10	0.08	0.51	–	–

Table S12. Post hoc permutation importance on the test period: drop in AUROC when a sensor group (top) or single sensor (bottom) is permuted. Negative values arise when the model's AUROC is below chance.

Group / sensor	Isolation Forest	MLP autoencoder	z-score
pressure	0.145	-0.413	0.162
thermal_current	-0.002	-0.501	0.002
digital	-0.002	-0.500	0.008
TP2	-0.001	-0.394	–
TP3	-0.001	-0.402	–
H1	-0.001	-0.389	–
DV_pressure	-0.001	-0.537	–

Reservoirs	-0.002	-0.395	–
Oil_temperature	-0.001	-0.515	–
Motor_current	-0.001	-0.386	–
COMP	-0.000	-0.428	–
DV_eletric	-0.000	-0.484	–
Towers	-0.000	-0.464	–
MPG	-0.001	-0.438	–
LPS	-0.001	-0.001	–
Pressure_switch	-0.000	0.003	–

Table S13. Stability of explanation rankings (Spearman correlation over sensors).

Comparison	Mean ρ	Minimum ρ	n
IF-SHAP seeds (F3, vs seed0)	0.88	0.84	4
IF-SHAP train_B vs train_A	0.95	0.95	1
IF-SHAP train_C vs train_A	0.73	0.73	1
MLP-AE error share seeds (F3, vs seed0)	0.76	0.65	2
IF-SHAP vs MLP-AE (F3)	0.16	0.16	1

S9. Computational cost

Fit time, inference time and model size are in Table S14 and Fig. S9. Scoring one window takes well under a millisecond for all detectors on two CPU cores.

Table S14. Fit time (s), inference time (ms per window) and serialized model size (kB) on two CPU cores.

Model	Fit (s)	Inference (ms)	Size (kB)	Features
Rolling z-score	0.000	0.0000	0.2	46
Isolation Forest	0.261	0.0076	1097.4	46
One-Class SVM	0.076	0.0153	76.5	46
MLP autoencoder	8.694	0.0004	14.0	46
LSTM autoencoder	14.880	0.0114	57.5	46

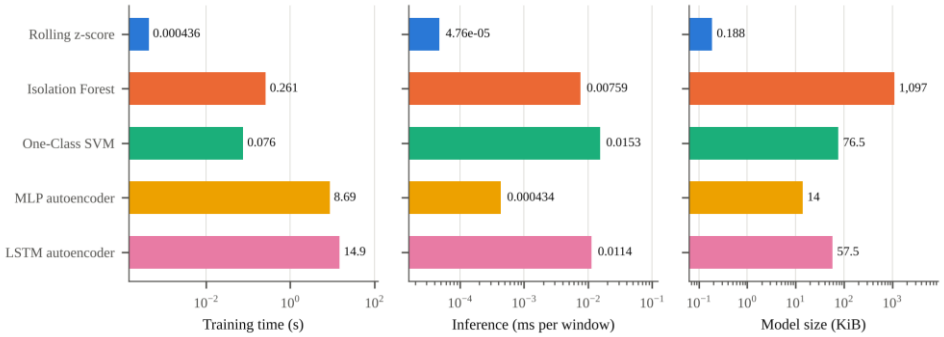


Figure S9. Fit time, inference time and model size per detector on two CPU cores (log scale).

S10. Related work

Table S15 summarizes the scope of the closest related studies. Protocols, data subsets and metrics differ, we did not re-run any of these methods, and we therefore compare scope rather than numbers.

Table S15. Scope of the closest related work on MetroPT data and explainable maintenance, as indicated by each cited record. Scope only; no numerical comparison.

Reference	Focus	Relation to this study
Davari et al. (2021)	Deep-learning anomaly detection for predictive maintenance of the APU	Same unit; we add chronological threshold fixing, event-level metrics and a contamination check
Barros et al. (2020)	Failure detection of the APU in an operational context	Operational framing; ours is offline and retrospective
Veloso et al. (2022)	Description of the MetroPT dataset	Source of the data and failure reports
Meira et al. (2023)	Data-driven predictive maintenance framework for railway systems	Broader framework; we examine one detection stage
Pashami et al. (2023)	Explainable predictive maintenance (perspective)	Motivates the explanation and stability analysis
Brito et al. (2021)	Explainable unsupervised fault detection, rotating machinery	Different asset; related explanation approach
Akshitha Mary et al. (2025)	Predictive modeling on MetroPT-3 (from the title)	Not re-implemented; protocol not compared

S11. Limitations and protocol history

Table S16 lists the limitations. Our first protocol used a contaminated validation set and unstable z-scores on near-constant channels. We archived it (folder results_v1 in the repository) and disclosed the revision in the main paper. The revised protocol was motivated by diagnostics that included test-period behavior, so the test results are not fully independent of design choices.

Table S16. Limitations of the study and what each means for the claims.

Limitation	Consequence
Partial data (28.8 % of rows; 5 February–30 April and the period after 26 June missing)	Failures 1 and 4 not evaluable; results hold for this subset only
Two events, both air leaks, one vehicle	No generalization claim; no significance tests; intervals are descriptive
Documented intervals approximate; maintenance date of failure 2 inferred	Delays and pre-onset zones uncertain by tens of minutes
Undocumented failure-like runs and digital-state windows in the normal class	False-alarm rates conservative; audit and strict results both reported
Protocol revised once after seeing test diagnostics (first version archived)	The evaluation is not a pristine single-pass test
Permutation importance and the contamination diagnostic use test data post hoc	Descriptive only; never used for model selection
Early warning not demonstrated	No claim beyond detection
Only four February days as a separate training set	Training-period comparisons are confounded by data volume

S12. Reproducibility

The full pipeline was run twice and every numeric result matched to within 1e-9 (timings excluded). Seven unit tests cover window causality, split disjointness, zone assignment, episode and threshold logic, detection delay and the state flag. The repository README lists the commands and the folder layout. Package versions are in requirements.txt.

References cited in the supplementary materials

[1] Akshitha Mary, A. C., Rakshinee, R., Jeyaseelan, S., Sakthivel, V. and Prakash, P., "Predictive modeling for metro performance using MetroPT3 dataset," Int. J. Adv. Comput. Sci. Appl. 16(8) (2025)

[2] Barros, M., Veloso, B., Pereira, P. M., Ribeiro, R. P. and Gama, J., "Failure detection of an air production unit in the operational context," in IoT Streams for Data-Driven Predictive Maintenance and IoT, Edge, and Mobile for Embedded Machine Learning, Springer, 61–74 (2020); https://doi.org/10.1007/978-3-030-66770-2_5

[3] Brito, L. C., Susto, G. A., Brito, J. N. and Duarte, M. A. V., "An explainable artificial intelligence approach for unsupervised fault detection and diagnosis in rotating machinery," arXiv:2102.11848 (2021)

[4] Davari, N., Veloso, B., Ribeiro, R. P., Pereira, P. M. and Gama, J., "Predictive maintenance based on anomaly detection using deep learning for air production unit in the railway industry," Proc. 8th IEEE Int. Conf. Data Sci. Adv. Anal. (DSAA), 1–10 (2021); https://doi.org/10.1109/DSAA53316.2021.9564181

[5] Meira, J., Veloso, B., Bolón-Canedo, V., Marreiros, G., Alonso-Betanzos, A. and Gama, J., "Data-driven predictive maintenance framework for railway systems," Intell. Data Anal. 27(4), 1087–1102 (2023); https://doi.org/10.3233/IDA-226811

[6] Pashami, S., Nowaczyk, S., Fan, Y., Jakubowski, J., Paiva, N., Davari, N., et al., "Explainable predictive maintenance," arXiv:2306.05120 (2023)

[7] Veloso, B., Ribeiro, R. P., Pereira, P. M. and Gama, J., "The MetroPT dataset for predictive maintenance," Sci. Data 9(1), 764 (2022); https://doi.org/10.1038/s41597-022-01877-3